INCH-POUND
MS21341H
19 October 2012
SUPERSEDING
MS21341G
15 December 1982

DETAIL SPECIFICATION SHEET

METER, TIME TOTALIZING, HERMETICALLY SEALED, 28 VDC

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-7793.

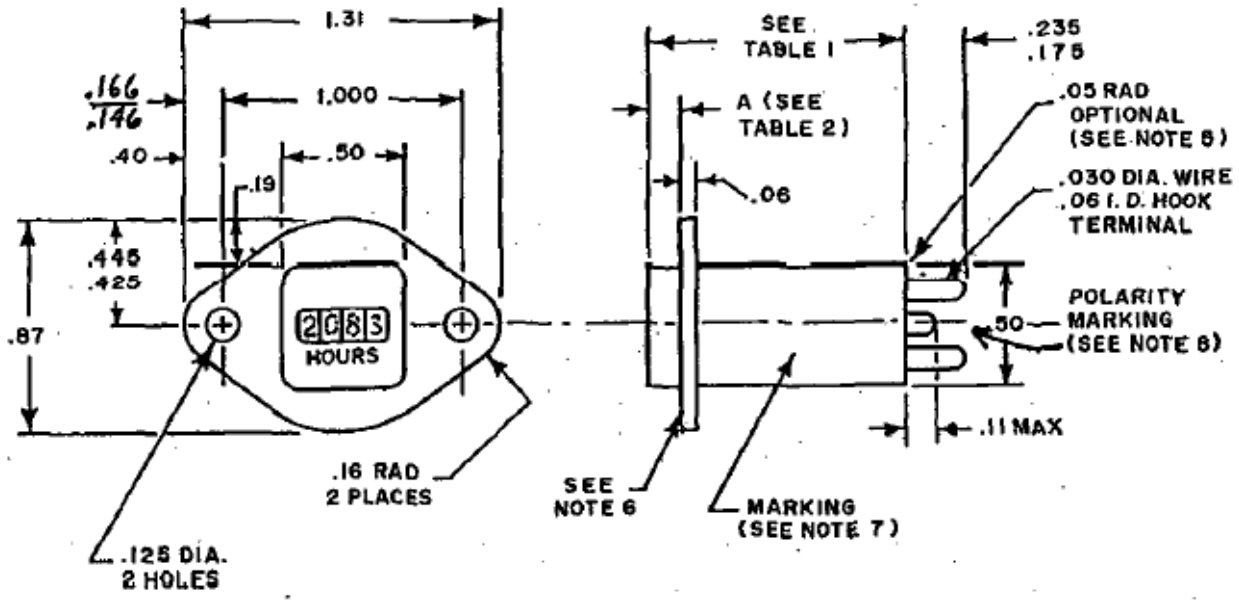



FIGURE 1. Meter configuration.



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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.02 for two place decimals and ±.005 for three place decimals
3. For design feature purposes, this specification sheet takes precedence over documents referenced herein.
4. Referenced documents shall be of the issue in effect on the date of invitation for bid.
5. Optional radius applies to all corners and edges.
6. Exposed tin plate on this surface of mounting flange only, per ASTM-B545, to provide an electrical ground for the case.
7. Marking surface is optional. Marking must not be obscured by mounting bracket.
8. Polarity marking (+) near positive terminal on this surface.

REQUIREMENTS:

1. Dimensions. See Figure 1 and Table I.

2. Electrical. Unless otherwise specified, all tests shall be at room pressure and temperature, approximately +23 °C.

3. Meter range. 0 to 9999 hours.

4. Scale designator. Hours

5. Mounting flange. Per Note 6 above, tin-plate .0002 to .0004-inch thick to provide means for electrical ground of the case (see Figure 1).

6. Transient protection. No temporary or permanent degradation shall be produced in the meter for the input voltage and time values shown on Figure 2 and Figure 3 (80V and 600V transient, respectively).

7. Ripple protection. The meter must maintain the accuracy specified in Table I when subjected to a cyclic peak of ripple voltage of less than 2.0 VDC and the frequency-voltage coordinates of Figure 4.

8. Operational check. The meter shall register accurate time over the input voltage range specified in Table I between -65 °C and +125 °C.

9. Case finish. The case shall be finished in a durable, dull or semi-gloss black color.



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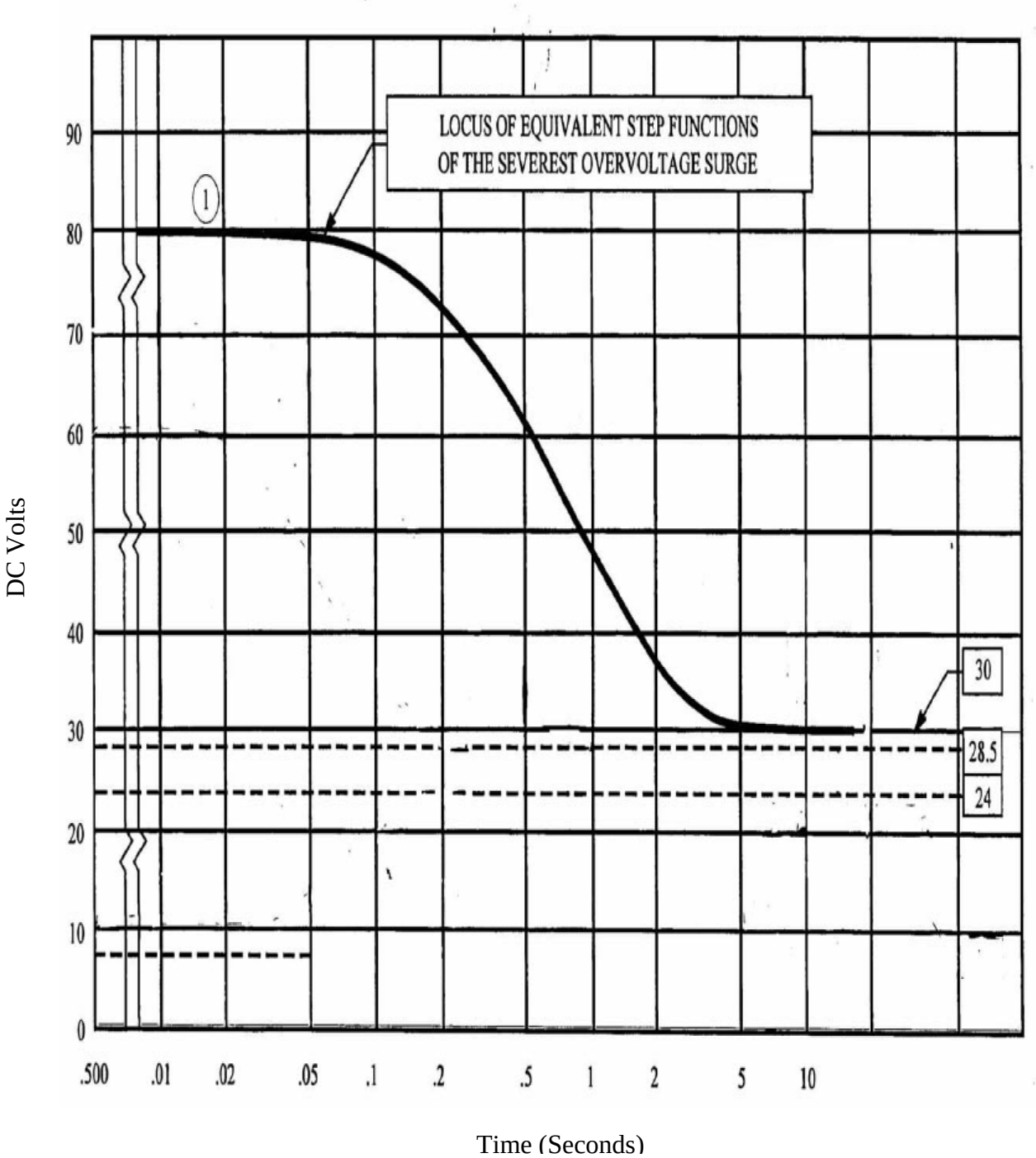



FIGURE 2. Transient surge step function loci limits.



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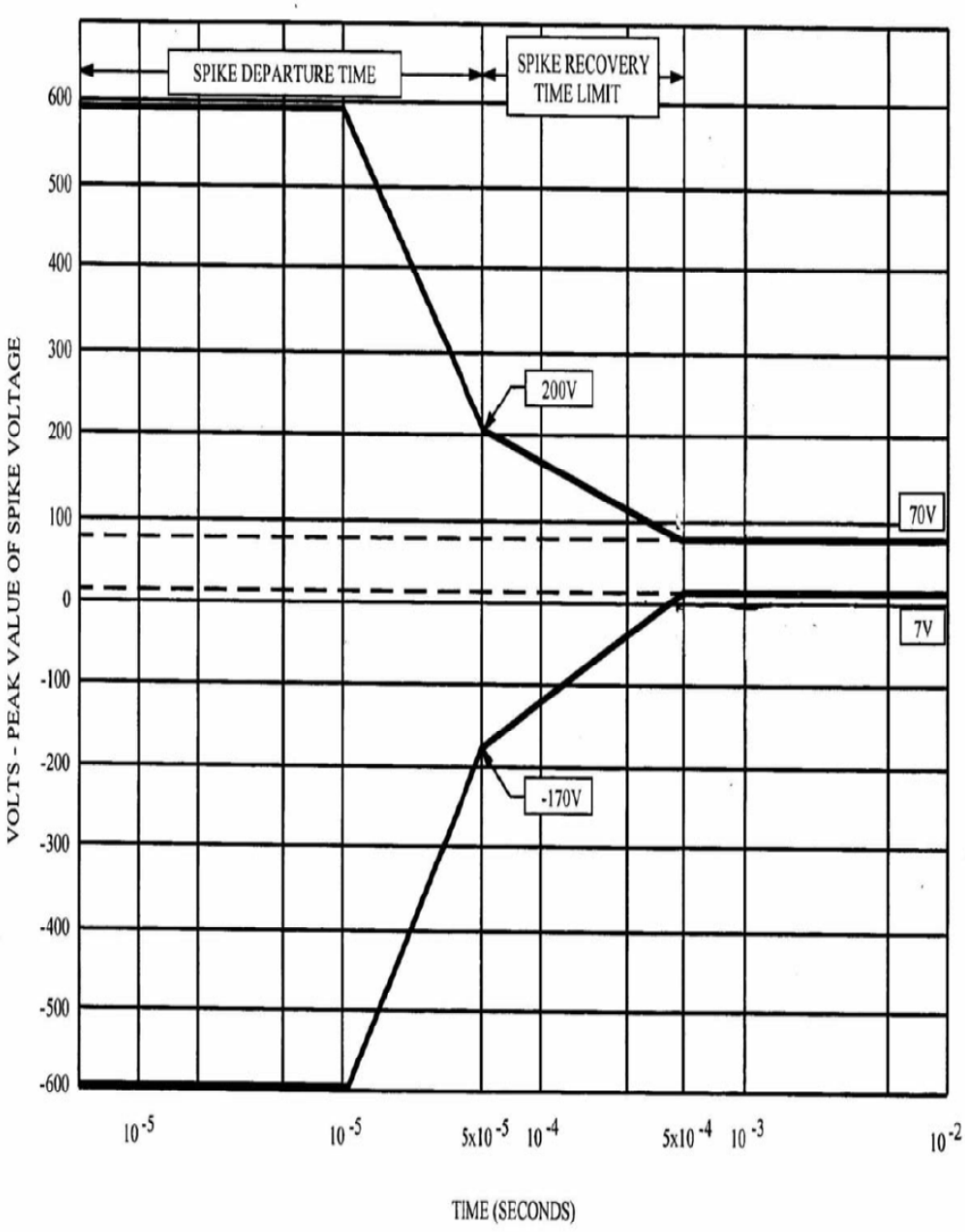



FIGURE 3. Envelope of spike voltages for direct current.



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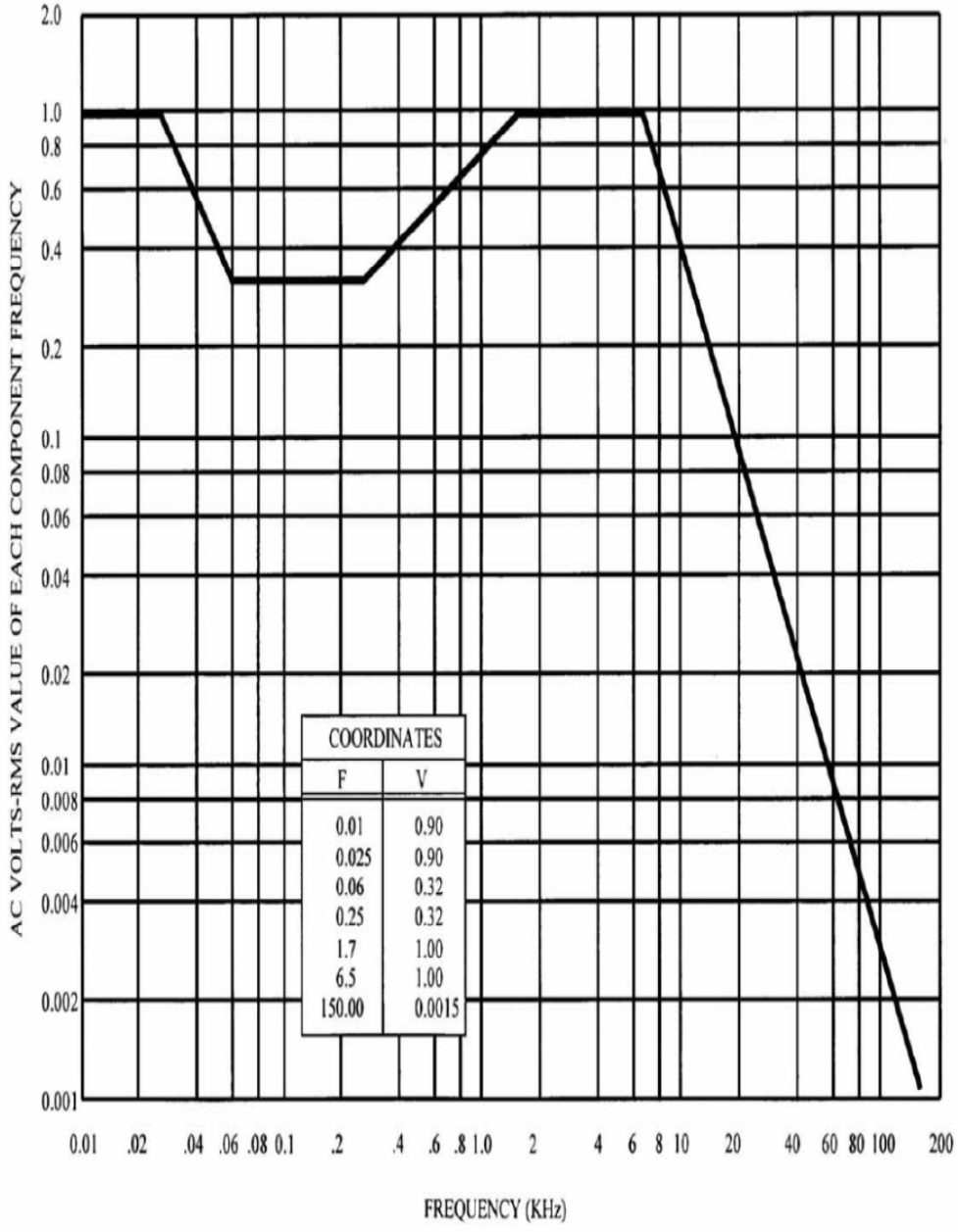



FIGURE 4. Frequency characteristics of ripple in 28 VDC electric systems.



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10. Marking. The meter shall remain clearly and legibly marked with the following information after subjection to the environmental requirements:

(a) Part number (Example: MS21341A-05)
(b) Source and date code per MIL-STD-1285.
(c) 28 VDC Nominal Voltage.

TABLE I. Meter configuration and requirements.

| Requirement | MS21341A | MS21341B |
|---|---|---|
| Case Length, Inch (See Figure 1) | 1.094/1.047 | 2.062/2.000 |
| Maximum Weight, Ounce | 0.6 | 1.5 |
| Input Voltage Range, VDC | 10 to 34 | 23 to 32 |
| Nominal Input Power, Watt | 0.085 (Average) | 1.000 |
| Maximum Input Power, Watt | 0.150 (Average) | 1.500 |
| Accuracy, Percent | 0.1 | 1.0 |

TABLE II. Mount setback dash numbers and dimensions (see Figure 1).

| Dash Number | Dim A ±.015 | Dash Number | Dim A ±.015 | Dash Number | Dim A ±.015 |
|---|---|---|---|---|---|
| -01 | .000 | -10 | .281 | -19 | .562 |
| -02 | .031 | -11 | .312 | -20 | .594 |
| -03 | .062 | -12 | .344 | -21 | .625 |
| -04 | .094 | -13 | .375 | -22 | .656 |
| -05 | .125 | -14 | .406 | -23 | .688 |
| -06 | .156 | -15 | .438 | -24 | .719 |
| -07 | .188 | -16 | .469 | -25 | .750 |
| -08 | .218 | -17 | .500 | -26 | .781 |
| -09 | .250 | -18 | .531 | | |

11. Conformance inspection. The following tests, run in the order shown on all meters, supersede the individual tests and sampling Plans A and B specified in MIL-DTL-7793. The maximum meter reading at the time of delivery shall be 0001.

(a) Examination of product.
(b) Operational shock shall consist of ten (10) operations for instant starting at ambient temperature, approximately 25 °C, five each at minimum and maximum input voltages specified in Table I of this specification sheet.
(c) Thermal shock per MIL-STD-202, Method 107, test condition B, modified as follows:

Test conditions:

1. During step 1, the meter shall be operated at its maximum input voltage as specified in Table I of this specification sheet.



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2. During step 3, the meter shall be operated at its minimum operating voltage as specified in Table I of this specification sheet.

MEASUREMENTS DURING CYCLING

1. The meter shall show no evidence of damage and shall not malfunction.

(d) Operation at room ambient (25 °C ±5 °C) and 28 VDC for 20 hours ±1 hour.
(e) Hermetic seal.
(f) Insulation resistance.
(g) Sea level dielectric stress (600V RMS, 2 to 5 seconds)
(h) All meters failing one or more of these conformance tests shall be rejected.

12. In addition to the above tests, three sample meters total (any part number) shall be subjected to the following tests every twenty four months.

(a) Salt spray.
(b) Vibrations.
(c) Shock.
(d) Life. Meters shall be operated at input voltage limits specified in Table I, for 100 hours in each of the following sequences:

(1) Maximum voltage at 125 °C.
(2) Minimum voltage at 125 °C.
(3) Maximum voltage at -65 °C.
(4) Minimum voltage at -65 °C.
(5) 28 VDC at room ambient.

(e) Voltage rise time as follows:

(1) Apply a voltage rising from zero to maximum input voltage in:
    (a) 3 milliseconds maximum
    (b) 2 ± .5 seconds
(2) Repeat both (a) and (b) tests five (5) times.
(3) The meter shall start in each case.

At the conclusion of these tests, meter readings shall be within the accuracy specified in Table I.

Qualification test.

The qualification tests are to be conducted in accordance with MIL-DTL-7793, except that four (4) sample meters (designated 1, 2, 3 and 4) shall be submitted to Group 1 tests per Table III of this specification sheet. Samples designated 1 and 2 shall be submitted to Group 2 test and samples designated 3 and 4 shall be submitted to Group 3 tests.



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TABLE III. Qualification tests.

| Test | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Group 1 | | |
| Examination of product | 3.8.1 | Visual examination (4.4.1) |
| | 3.7.2 | |
| | 3.7.3 | |
| | 3.7.6 | |
| | 3.9 | |
| Operational check | 3.8.2 | 4.4.2 |
| Group 2 | | |
| Power supply tolerance | 3.8.4 | 4.4.4 |
| Insulation resistance | 3.8.5 | 4.4.5 |
| Dielectric strength | 3.8.6 | 4.4.6 |
| Altitude | 3.8.7 | 4.4.7 |
| Thermal shock | 3.8.8 | 4.4.8 |
| Moisture resistance | 3.8.9 | 4.4.9 |
| Salt spray | 3.8.10 | 4.4.10 |
| Vibration | 3.8.11 | 4.4.11 |
| Shock | 3.8.12 | 4.4.12 |
| Hermetic seal | 3.8.13 | 4.4.13.1 |
| Group 3 | | |
| Life accuracy | 3.8.14 | 4.4.14 |
| Electromagnetic compatibility | 3.8.15 | 4.4.15 |



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6. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-7793, this specification sheet references MIL-STD-1285, MIL-STD-202, and ASTM-B545.

Custodians:
Navy - AS
Air Force - 99

Preparing activity:
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Project 6645-2012-020

Review activity:
DLA - GS1

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